

VEGABrain: a Visualizer of Enriched Gene Analysis for the developing human brain

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Abstract

RNA sequencing datasets are widely reused resources in developmental neuroscience, yet inspecting them spatially remains unclear and difficult. Here, we present VEGABrain, an openly accessible web tool that renders bulk RNA-seq onto brain cortical and subcortical atlas surfaces across five human developmental windows spanning the first trimester to adulthood. Beyond single-gene maps, VEGABrain contributes a composite per-region enrichment layer using single-sample gene set enrichment analysis (ssGSEA) scores precomputed for ~19,000 gene sets drawn from Gene Ontology, Reactome, KEGG, BioCarta, the Human Phenotype Ontology and the MSigDB hallmark collection, and aggregated to a score for every gene set, developmental window and brain region. Furthermore, VEGABrain allows users to submit a custom gene list and obtain its ssGSEA map across development, computed on demand. VEGABrain is open source and requires no installation, offering easy-to-query expression throughout development and brain areas.

Keywords: Neuroinformatics, Brain development, BrainSpan, Gene set enrichment, ssGSEA, Data visualization, Web application

1 Introduction

The transcriptome of the developing human brain is one of the most heavily reused datasets in neuroscience. The BrainSpan atlas (BrainSpan 2011; Kang et al. 2011; Miller et al. 2014) sampled bulk tissue from cortical and subcortical structures across prenatal and postnatal donors, and subsequent integrative work built on it to relate developmental expression to neuropsychiatric risk (Li et al. 2018) and to evolutionary divergence between human and macaque (Zhu et al. 2018).

What has not kept pace is the ease of seeing it. A researcher who wants to know where and when a gene, or a set of genes, is expressed in the developing brain has essentially two options today, and both impose costs that are unrelated to the biological question.

The first is the resource portal. Portals in this family—BrainSpan itself, and the wider family of Allen Institute browsers (Lau et al. 2008; Sunkin et al. 2012)—return the data as tables, line plots or heat maps indexed by structure acronym. This, unfortunately, is not anatomical: the reader must hold the mapping from DFC, MFC or STC to a location in the brain, and a heat map of 26 structure codes does not convey a spatial gradient the way a rendered cortex does. It is also gene-at-a-time. There is no route from a gene set, a pathway, or the output of a differential-expression analysis to a spatial view. Newer hosted portals have narrowed part of this gap: BrainScope links genes and samples through t-SNE maps to choropleths over anatomical slices (Huisman et al. 2017), and SMDB browses spatial transcriptomic data in two and three dimensions (Cao et al. 2023). Neither, however, scores a gene set across a human developmental parcellation (Section 4.1).

The second option is to do it yourself. Excellent programmatic brain-plotting tools exist—`ggseg` and `ggseg3d` (Mowinckel and Vidal-Piñeiro 2020) and `cerebroViz` (Bahl et al. 2017) in R, `neuromaps` in Python (Markello et al. 2022), and `abagen` for the adult Allen Human Brain Atlas (Hawrylycz et al. 2012; Markello et al. 2021). But that route presupposes a user who writes R or Python, who is willing to download and hold a $52,376 \times 524$ expression matrix locally, and who will construct and validate the mapping from BrainSpan’s tissue nomenclature onto an atlas parcellation. Each of those steps is a place where a project stalls or a mapping silently diverges between laboratories (Markello et al. 2021).

Additionally, modern analyses rarely end with one gene. They end with a list, be it a differentially expressed set, a module from a co-expression network or a panel of risk genes for a disorder. The natural question is then not “where is gene X expressed?” but “where and when is this programme active?”. Answering it requires collapsing many genes into a single per-sample summary. Single-sample gene set enrichment analysis (ssGSEA) (Barbie et al. 2009), a per-sample variant of GSEA (Subramanian et al. 2005) implemented alongside GSVA (Hänzelmann et al. 2013), provides exactly such a composite score, and has been used as a quantitative readout of pathway activity in expression data (Yi et al. 2020). The incorporation of composite multigene activity scores (such as those generated by ssGSEA or GSVA) into anatomical models directly addresses the challenge of condensing high-dimensional expression profiles into interpretable summaries, ensuring noise reduction and dimensionality reduction

compared to classical single-gene-based analyses (Hänzelmann et al. 2013). To our knowledge no tool has applied it as a *spatial* layer over the developing human brain.

VEGABrain (Visualizer of Enriched Gene Analysis) addresses these three gaps in one interface, reachable at <https://mcblab.github.io/BrainMap/> with no installation, account or download. It maps BrainSpan samples onto cortical and subcortical atlas parcels across five developmental windows; it adds a composite enrichment layer of ssGSEA scores precomputed for 18,650 gene sets from the Gene Ontology (Ashburner et al. 2000; The Gene Ontology Consortium et al. 2023), Reactome (Gillespie et al. 2022), KEGG (Kanehisa 2000), BioCarta, the Human Phenotype Ontology (Köhler et al. 2021) and the MSigDB hallmark collection (Liberzon et al. 2011, 2015); and it scores user-submitted gene lists on demand, closing the loop from an enrichment result (Chen et al. 2013; Kuleshov et al. 2016) back to anatomy. A submitted list may also carry log₂ fold-change values from an external comparison—a different disease model, treatment or condition—which are then mapped against the BrainSpan expression data, so that analyses can follow directions the dataset was not originally built for, not just the comparisons already included in it. Both the maps and the numbers behind them are exportable, so that VEGABrain is a starting point for downstream analysis, consistent with the FAIR principles for reusable research data (Wilkinson et al. 2016).

2 Materials and Methods

2.1 Expression data and preprocessing

VEGABrain is built on the BrainSpan Developmental Transcriptome RNA-Seq release, Gencode v10 RPKM values summarised to genes (BrainSpan 2011; Miller et al. 2014). The distributed matrix contains 52,376 gene rows and 524 sample columns. Rows with duplicate HGNC symbols are dropped, keeping the first occurrence, and values are transformed as $\log_2(\text{RPKM} + 1)$. No further normalisation or batch correction is applied; the released RPKM values are used as the provider distributes them, so that scores computed here remain comparable with analyses performed directly on the public matrix. Preprocessing and the scoring described in Section 2.4 are performed once, offline (Figure 1).

2.2 Developmental windows

BrainSpan donors span 31 distinct ages from 8 post-conceptual weeks to 40 years. Individual ages are too sparsely sampled to render as separate maps, so donors are binned into five developmental windows: first trimester ($n = 5$ donors), second trimester ($n = 10$), third trimester ($n = 5$), infant ($n = 8$) and adult ($n = 14$). The donor counts are also given in Table 1; the donor count is displayed in every facet label in the interface so that the reader is never shown a map without also being shown how much evidence stands behind it.

2.3 Mapping BrainSpan structures to an atlas

BrainSpan labels tissue by dissected structure (for example *dorsolateral prefrontal cortex*), whereas **ggseg** draws atlas parcels (for example *rostral middle frontal*). Cortical

regions map onto the Desikan–Killiany parcellation (Desikan et al. 2006) and subcortical structures onto the FreeSurfer `aseg` segmentation (Fischl et al. 2002), both accessed through `ggseg` (Mowinckel and Vidal-Piñeiro 2020).

Each atlas parcel is assigned a single BrainSpan structure. BrainSpan samples far fewer structures than the atlases draw, so the assignment is many-to-one: 22 BrainSpan structures cover 42 atlas regions, and parcels that fall within one dissected structure share its value (for example, the rostral and caudal middle frontal parcels both take the dorsolateral prefrontal cortex samples). Atlas regions with no BrainSpan counterpart, such as the brain stem, pallidum and choroid plexus, are left unassigned. The complete assignment is a single table distributed with the source code, so it can be inspected and changed rather than reimplemented per project.

The interface exposes two levels of granularity. At the *micro* scale each of the 42 mapped atlas regions carries its own value. At the *macro* scale regions are pooled into six anatomical groupings—frontal, parietal, temporal and occipital lobes, subcortical structures and cerebellum—the mean is taken within each grouping, and that mean is painted back onto every region belonging to it. The macro scale exists because BrainSpan sampling density varies sharply across structures: pooling trades spatial resolution for donor support, and letting the user switch between the two makes that trade-off explicit rather than hidden. Regions with no corresponding sample are drawn in grey rather than imputed.

Every map is rendered as four stacked panels—right lateral cortex, right medial cortex, subcortical sagittal and subcortical coronal—each faceted into the five developmental windows, giving a single figure in which spatial and temporal structure are read on orthogonal axes.

2.4 Composite enrichment scores

For a gene set G and a sample j , ssGSEA (Barbie et al. 2009) ranks all genes by their expression within that sample and computes an enrichment score from the normalised difference between the empirical cumulative distribution of the genes in G and that of the genes outside it:

$$\text{ES}(G, j) = \sum_{i=1}^N \left(P_w^G(G, j, i) - P^{NG}(G, j, i) \right), \quad (1)$$

where i indexes genes in decreasing order of expression in sample j , P_w^G is the weighted cumulative distribution over genes in the set and P^{NG} the cumulative distribution over the rest. Because the score is computed from within-sample ranks, it is invariant to the monotone transformation applied in Section 2.1 and does not require a reference group or a phenotype contrast—which is what makes it usable here, where each sample is a single donor–structure pair rather than a member of a case/control design. We use the `ssgseaParam` interface of the `GSVA` package (Hänzelmann et al. 2013) with default parameters.

Gene sets are retrieved through `msigdbR` (Dolgalev 2025) for *Homo sapiens* and comprise the MSigDB hallmark collection (Liberzon et al. 2015), the canonical pathway subcollections BioCarta, KEGG (Kanehisa 2000) and Reactome (Gillespie et al.

2022), and the full C5 ontology collection, which supplies Gene Ontology biological process, molecular function and cellular component terms (Ashburner et al. 2000; The Gene Ontology Consortium et al. 2023) together with Human Phenotype Ontology terms (Köhler et al. 2021). The resulting 18,650 sets are broken down by collection in Table 1.

Scores are computed once for every gene set against all 524 samples, then aggregated: for each (gene set, developmental window, region) triple the mean score over the contributing samples is stored, and analogously for macro-regions, so that an ontology query needs no enrichment run at request time. When instead a user submits their own gene list, the symbols are validated against the matrix (unmatched symbols are reported back rather than silently dropped), ssGSEA is run for that single set across all samples, and the same aggregation is applied before rendering.

2.5 Implementation

VEGABrain is an R (R Core Team 2026) service built with `plumber` (Schloerke and Allen 2024) behind a browser interface (Figure 1). Maps are rendered with `ggplot2` and `ggseg` (Mowinckel and Vidal-Piñero 2020), and can be downloaded as vector (SVG) figures together with the per-region values behind them as CSV. The same service is exposed as a public HTTP API, documented in the source repository, so that maps and tables can also be retrieved programmatically.

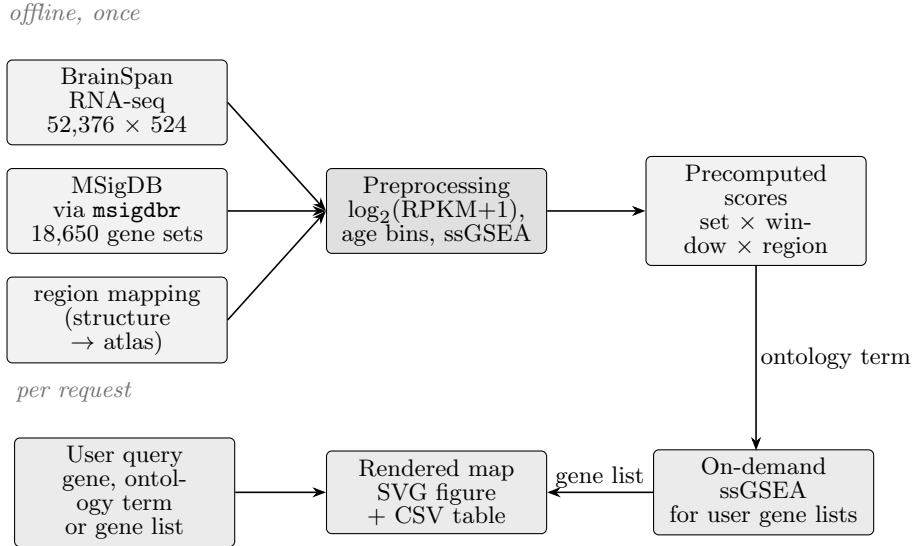


Fig. 1 VEGABrain data flow. The top row runs once, offline; the bottom row runs per request. Ontology queries are served from the precomputed score table, whereas a user-submitted gene list triggers a fresh ssGSEA run over all 524 samples.

3 Results

VEGABrain offers three query modes—a single gene, a precomputed ontology term, or a user-submitted gene list—on a single working page. All three produce the same four-panel, five-window figure at either region granularity (Section 2.3), and a reference map labelled by region name is available so that unfamiliar atlas parcels can be identified. Every map can be exported together with the table of values it was painted from, so a figure produced in VEGABrain and a statistic computed from its export cannot disagree. We illustrate each mode with a case whose developmental behaviour is independently established, so that the maps can also be judged for biological face validity.

3.1 Single-gene maps

Gene mode renders the mean $\log_2(\text{RPKM} + 1)$ of any gene in the matrix per region and developmental window, on a fixed colour scale bounded at 6, so that maps for different genes are directly comparable rather than each being auto-scaled to its own range. Oligodendrocyte and myelination markers (*SOX10*, *MBP*, *PLP1*) should be low prenatally and rise sharply in the infant and adult windows as myelination proceeds; Figure 2 shows *SOX10*. [TODO: report the observed per-window means].

3.2 Gene-set maps

Ontology mode returns the mean ssGSEA score of any of the 18,650 precomputed gene sets per region and window (Figure 3). This is the composite layer described in Section 2.4, and it is what distinguishes VEGABrain from a gene-at-a-time browser: a single map summarises the coordinated behaviour of tens to hundreds of genes. The member genes of the selected set are listed alongside the map, so the composition of any score is inspectable. Neurogenesis-associated terms such as GOBP_FOREBRAIN_GENERATION_OF_NEURONS should peak in the prenatal windows and decline postnatally; [TODO]. The composite score is informative precisely here: the individual genes of a neurogenesis term are noisy across 42 sparsely sampled regions, whereas the ssGSEA summary of the term is not.

3.3 User-submitted gene lists

In gene-list mode, a pasted list of symbols is validated and scored on demand (Figure 4). Symbols absent from the matrix are listed back to the user explicitly, so a map is never quietly computed on a subset of what was submitted. This is the path from an external result to anatomy: the output of a differential expression analysis, a co-expression module, or a disorder risk panel can be mapped across development directly, without the user preparing data or writing code. GABAergic interneuron markers (*GAD1*, *GAD2*, *SLC32A1*) should be present from the second trimester onward with subcortical and then cortical distribution; [TODO].

A further case tests the on-demand path against a model whose developmental timing is experimentally fixed rather than merely presumed. Fischer et al. (2021) exposed human pluripotent stem cells to chronic intermittent ethanol throughout a

50-day differentiation to cortical neurons, which corresponds to the first trimester, and report an upregulated dorsal/glutamatergic specification programme. Submitting those 13 symbols (Figure 4) through gene-list mode returns a map whose region-averaged score declines monotonically across the five windows—0.81, 0.74, 0.58, 0.47, 0.44—so the signature is maximal in exactly the window the *in vitro* model represents, and the ordering is recovered from BrainSpan alone, with no information from the original experiment. Because the exposure window is part of the experimental design, this is a stronger test than a disease with a merely prenatal attribution.

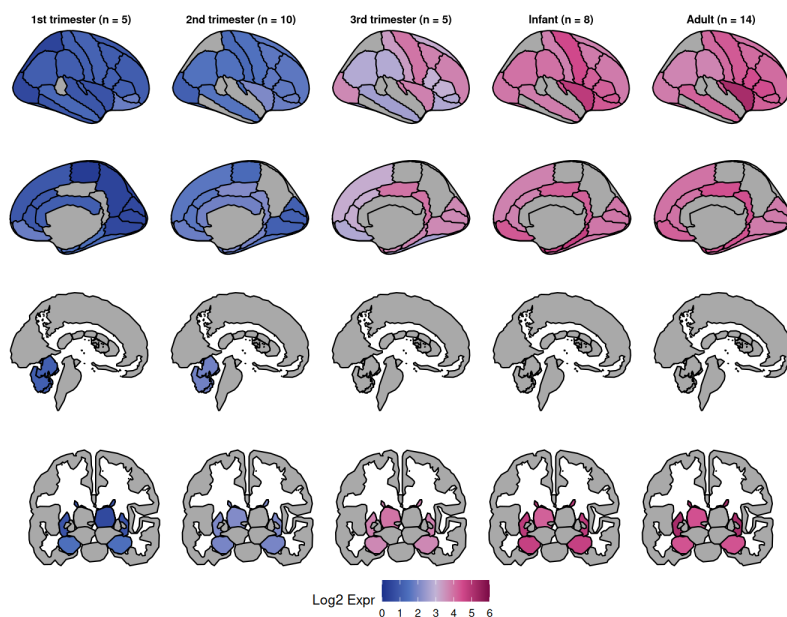


Fig. 2 Single-gene mode. Mean $\log_2(\text{RPKM} + 1)$ for *SOX10* across the five developmental windows (columns) in four views (rows): right lateral cortex and right medial cortex on the Desikan–Killiany parcellation, and subcortical structures in sagittal and coronal section on the *aseg* segmentation. Grey indicates regions with no BrainSpan sample in that window.

4 Discussion

VEGABrain lowers the cost of asking a spatial question of the developing human transcriptome from a small project to a few seconds in a browser. The composite enrichment layer extends the same reasoning to the unit that modern analyses actually produce—a gene set—and the on-demand mode means a user’s own list is a first-class input rather than something the tool’s authors had to anticipate.

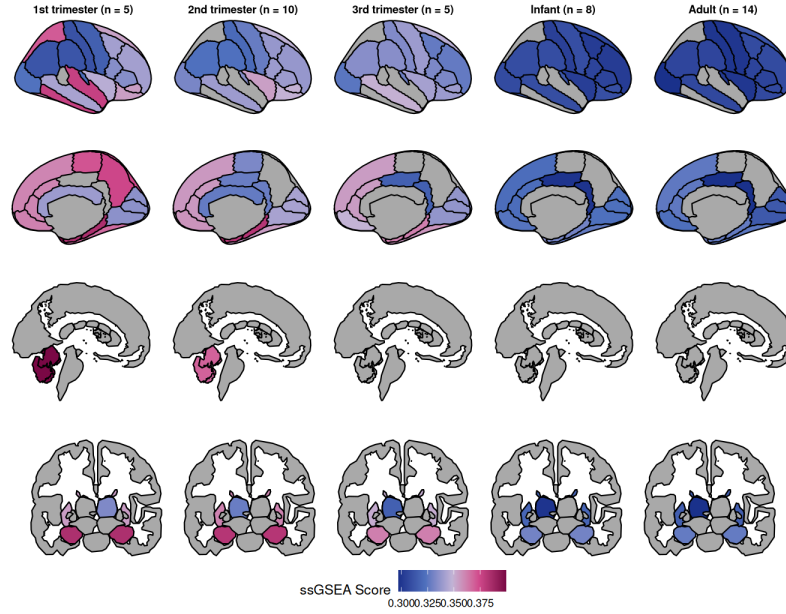


Fig. 3 Ontology mode: composite ssGSEA score for `GOBP_FOREBRAIN_GENERATION_OF_NEURONS` across development. The score summarises every gene in the term into one value per region and window; the same term can be viewed at micro granularity (42 atlas regions) or macro granularity (six anatomical groupings).

Table 1 Data underlying VEGABrain.

Property	Value
Expression source	BrainSpan RNA-Seq, Gencode v10 RPKM, gene-summarised
Matrix dimensions	52,376 genes $\times$ 524 samples
Transformation	$\log_2(\text{RPKM} + 1)$; duplicate symbols dropped
Developmental windows	5 (see below)
1st trimester	5 donors
2nd trimester	10 donors
3rd trimester	5 donors
Infant	8 donors
Adult	14 donors
Atlas regions (micro)	42 (Desikan–Killiany cortical + <code>aseg</code> subcortical)
Anatomical groups (macro)	6 (frontal, parietal, temporal, occipital, subcortical, cerebellum)
Views per map	4 (DK lateral, DK medial, <code>aseg</code> sagittal, <code>aseg</code> coronal)
Gene sets scored	18,650 total
GO biological process	7,538
Human Phenotype Ontology	5,793
GO molecular function	1,872
Reactome	1,839
GO cellular component	1,080
BioCarta	292
KEGG (legacy)	186
MSigDB hallmark	50

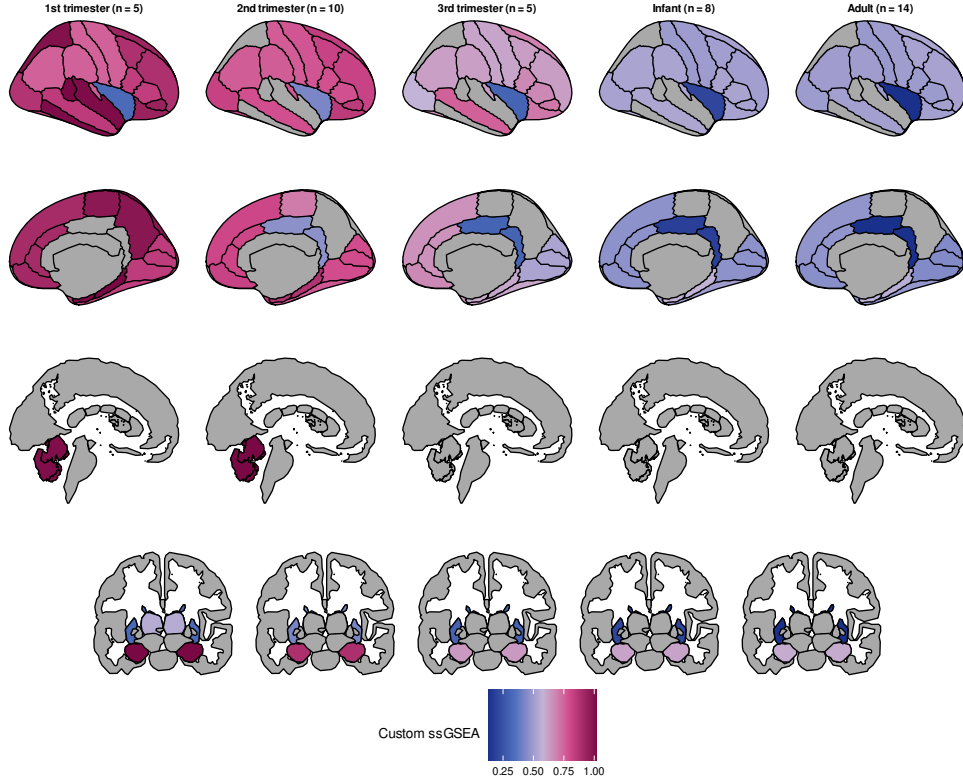


Fig. 4 User-submitted gene-list mode. ssGSEA computed on demand for the dorsal/glutamatergic specification programme that [Fischer et al. \(2021\)](#) report as upregulated by chronic intermittent ethanol exposure in human pluripotent stem cells differentiated to first-trimester-equivalent cortical neurons (*EMX2*, *LHX1/2/5/9*, *OTX2*, *FEZF2*, *NEUROD1/2/6*, *NEUROG2*, *TBR1*, *EOMES*). The submitted list is a published differential-expression result, not a curated marker panel, and it is mapped without local installation or coding. The score is highest in the cortical parcels of the first two prenatal windows—the period the *in vitro* model corresponds to—and falls monotonically thereafter, while the striatum stays low throughout, recovering the dorsal–ventral contrast the signature encodes.

4.1 Relation to existing tools

Table 2 positions VEGABrain against the tools a researcher would otherwise reach for. The comparison is deliberately along axes of *what the user must supply*, not of raw capability: each of these tools is stronger than VEGABrain in its own domain, and the point is that none of them closes the particular path VEGABrain is built for—from a gene set to a scored, parcellated map of the developing brain, in a browser.

As noted in the Introduction, the portals are gene-at-a-time and the programmatic tools take a value per region as *input*, leaving the user to obtain, preprocess and map the developmental expression data themselves. *cerebroViz* ([Bahl et al. 2017](#)) is the closest of them in intent, having been written for spatiotemporal brain data specifically and shipping region vocabularies for BrainSpan, GTEx and Roadmap Epigenomics; it nonetheless remains a drawing layer over a matrix the user provides, carries no

Table 2 VEGABrain relative to comparable tools. ✓ = supported, — = not supported or out of scope.

	Developmental axis	Anatomical rendering	Gene-set scoring	Custom list input	No install required
VEGABrain (this work)	✓	✓	✓	✓	✓
BrainSpan portal (BrainSpan 2011)	✓	—	—	—	✓
Allen browsers (Lau et al. 2008; Sunkin et al. 2012)	—	✓	—	—	✓
BrainScope (Huisman et al. 2017)	✓	✓	—	✓	✓
SMDB (Cao et al. 2023)	—	✓	—	—	✓
ggseg (Mowinckel and Vidal-Piñeiro 2020)	—	✓	—	✓	—
neuromaps (Markello et al. 2022)	—	✓	—	✓	—
abagen (Markello et al. 2021)	—	✓	—	✓	—
cerebroViz (Bahl et al. 2017)	✓	✓	—	✓	—
Programmatic plotting guides (Chopra et al. 2023)	—	✓	—	✓	—
Enrichr (Chen et al. 2013; Kuleshov et al. 2016)	—	—	✓	✓	✓
VoxHunt (Fleck et al. 2021)	✓	✓	—	✓	—

“Developmental axis” means the tool natively presents variation across prenatal-to-postnatal time. “Anatomical rendering” means values are drawn onto a picture of the brain, in whatever form the tool adopts—a parcellated surface, a slice choropleth or a three-dimensional point cloud. “Gene-set scoring” means it produces a composite per-sample score for a set of genes rather than plotting genes individually; a tool that displays the mean expression of a set, or that hands the set to an external enrichment server, does not qualify. “Custom list input” means the user may supply their own genes or their own per-region values.

expression data of its own, and computes no enrichment. The same qualification applies to the code-first route as a whole. Chopra et al. (2023) argue for it persuasively and lower its cost with a survey of the available packages and a generator for template plotting code, but a guide to writing the code does not remove the prerequisites the code still has: a working R or Python environment, a local copy of the expression matrix, and a mapping from tissue nomenclature onto a parcellation.

Among hosted resources, BrainScope (Huisman et al. 2017) is the nearest neighbour to VEGABrain, and the only other entry in Table 2 that is at once web-based, developmental and willing to take a user’s own gene set. The abstraction differs, and with it the question answered. BrainScope arranges the Allen adult and developing atlases as linked t-SNE maps of genes and samples, localising samples anatomically through choropleths over selected slices, and summarises a submitted set by its mean expression while delegating enrichment to Enrichr and ToppGene. VEGABrain asks the narrower question of where a set sits in a named parcellation, window by window, and answers it with a rank-based composite score rather than a mean, drawn on the Desikan–Killiany and aseg atlases that the imaging literature already uses. SMDB (Cao et al. 2023) is likewise hosted and likewise anatomical, but for a different data type and species: it browses spatial transcriptomics sections against Allen’s mouse brain coordinate framework and reconstructs morphology in three dimensions, with no human developmental axis and no gene-set layer.

Enrichr (Chen et al. 2013; Kuleshov et al. 2016) performs the gene-set reasoning but returns ranked term tables, with no anatomical or developmental dimension. VoxHunt (Fleck et al. 2021) maps expression signatures onto developmental reference atlases,

primarily to assign identity to *in vitro* organoid samples, and is an R package rather than a hosted service. VEGABrain occupies the intersection the others leave open: hosted, developmental, parcellated, and gene-set *scored*.

4.2 Limitations

Several limitations bound the interpretation of any map produced here, and the interface is designed to keep them visible rather than to smooth them over. First, donor support is small and uneven: each developmental window rests on 5 to 14 donors, and not every structure is sampled in every window. Donor counts are printed in every facet label and unsampled regions are drawn in grey rather than interpolated, but a difference between two regions in a single window should be read as descriptive, not as a test. The many-to-one assignment of BrainSpan structures to atlas parcels (Section 2.3) likewise means that neighbouring parcels sharing a structure cannot differ.

Second, the data are bulk tissue. BrainSpan samples are dissected tissue, so a regional value reflects both cell-intrinsic expression and cell-type composition: a rise in an oligodendrocyte programme across development is as consistent with more oligodendrocytes as with more transcription per cell. Single-cell atlases resolve this and are the natural next data source.

Third, ssGSEA scores are built from within-sample ranks. Equation 1 yields a value that is comparable across samples for a fixed gene set, but not across gene sets of different sizes and not on an absolute scale. Colour scales for ontology maps are therefore free rather than fixed, and a score should be read as a spatiotemporal pattern, not as a magnitude.

Future work follows directly from these bounds: incorporating single-cell and single-nucleus developmental atlases, adding statistical contrast between windows or regions rather than descriptive display alone and allowing user-supplied parcellations.

5 Conclusion

VEGABrain makes the developmental human brain transcriptome, and the activity of gene sets within it, visible anatomically without installation, coding or data preparation. Maps and their underlying tables are exportable, the structure-to-atlas mapping is published rather than implied, and the source is open. The tool is available at <https://mcblab.github.io/BrainMap/>.

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Declarations

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Competing interests. The authors declare no competing interests.

Ethics approval and consent to participate. Not applicable. This study analyses publicly released, de-identified data from the BrainSpan Developmental Transcriptome and involved no new human subjects.

Data availability. The expression data analysed here are the BrainSpan Developmental Transcriptome RNA-Seq Gencode v10 summarised-to-genes release, publicly available at <https://www.brainspan.org>. Gene sets were obtained from MSigDB through the `msigdb` R package. The derived per-region, per-window score tables served by VEGABrain are downloadable as CSV directly from the application.

Code availability. VEGABrain is open source at <https://github.com/MCBLab/BrainMap>. The application is deployed at <https://mcblab.github.io/BrainMap/>; its public HTTP API is documented in the repository.

Author contributions. IM: Formal analysis, Methodology, Visualization, Writing – review and editing. JVFC: Formal analysis, Writing – review and editing. DMC: Conceptualization, Data curation, Investigation, Supervision, Writing – original draft, Writing – review and editing.

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